PONTSHO MBIZO

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SUMMARY

Graduate Electrical & Computer Engineer with four industry placements across defence, mining, and infrastructure. I combine rigorous quantitative analysis with structured problem-solving, possessing a track record of delivering parameterised tools and embedded systems that engineering teams actively adopt. I communicate complex technical findings clearly to diverse stakeholders and am driven to build solutions where sharp thinking translates into tangible, operational outcomes.

WORK EXPERIENCE

Project Engineering Intern, Tharisa Minerals *Dec 2025 – Jan 2026*

- Built a parameterised three-phase MCC sizing tool in Excel: full-load currents, voltage drop, power factor, and operating cost per motor — validated analytically and against manufacturer data (SANS 1507, WEG IE3 motors).
- Built-in percentage-deviation check surfaces specification errors before procurement; methodology report accepted by engineering department.
- Tool adopted by the site team as the baseline for all feeder studies; presented cost-comparison results to the project engineering team.

Engineering Intern, Cybicom Atlas Defence *Jun – Jul 2025*

- Designed a high-speed FPGA capture system (Verilog, Intel MAX10) to non-destructively extract mission-critical data from battery-backed SRAM on submarine hardware — 200 MHz PLL, 70 ns timing precision.
- Built a bidirectional, multi-threaded UDP communication system in C (POSIX threads) for a motion simulator, with custom checksums and full handover documentation.
- Designed and 3D-printed a parametric mounting system in Fusion 360 to secure a 4U simulation computer against inertial loads inside a simulator instructor station.
- Participated in an Acceptance Test Procedure for a full motion simulation system across mechanical, electrical, and software subsystems.

Network Engineer/Artisan (Vacation Work), Tharisa Minerals *Dec 2024 – Jan 2025*

- Restored LAN connectivity across multiple plant sites; investigated Microsoft Cloud PC as a cost-reduction strategy for remote operations.
- Expanded network infrastructure by connecting new office spaces and improving plant-wide network access.

Electrical and Instrumentation Vacation Work, Tharisa Minerals *Dec 2023 – Jan 2024*

- Assisted Electrical Artisans and Instrumentation Technicians in routine maintenance, inspection, and upkeep of electrical equipment and control systems.

EDUCATION

BSc Eng Electrical and Computer Engineering *2022 – Present*

University of Cape Town

- Final-year: High Performance Embedded Systems, DSP & ML, Controls & Instrumentation, Law for Engineers; Design project — oceanographic sensor buoy (full systems engineering lifecycle).
- Earlier coursework: Embedded Systems, Electronic Devices & Circuits, Control Systems, Programming.

ADDITIONAL INFORMATION

- **Software & Data:** C/C++, Python (NumPy, pandas, scikit-learn), MATLAB
- **Hardware & Firmware:** Verilog HDL, KiCAD, Intel Quartus Prime
- **Design & Tools:** Fusion 360, Git, Linux (Fedora), LaTeX, MS Excel
- **Languages:** English (Fluent), Setswana (Native), isiZulu (Basic), isiXhosa (Basic)
- **Driver's License:** None
- **Awards / Activities:**
  1. UCT Plus Silver — 45 Hours Role Modelling Leadership Award for residence mentoring.
  2. Dell Young Leader.
  3. Electrical Engineering peer mentor at UCT.
- *References available upon request.*